

PROJECT DESIGN PHASE

Solution Architecture

Date	01 JULY 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	2 Marks

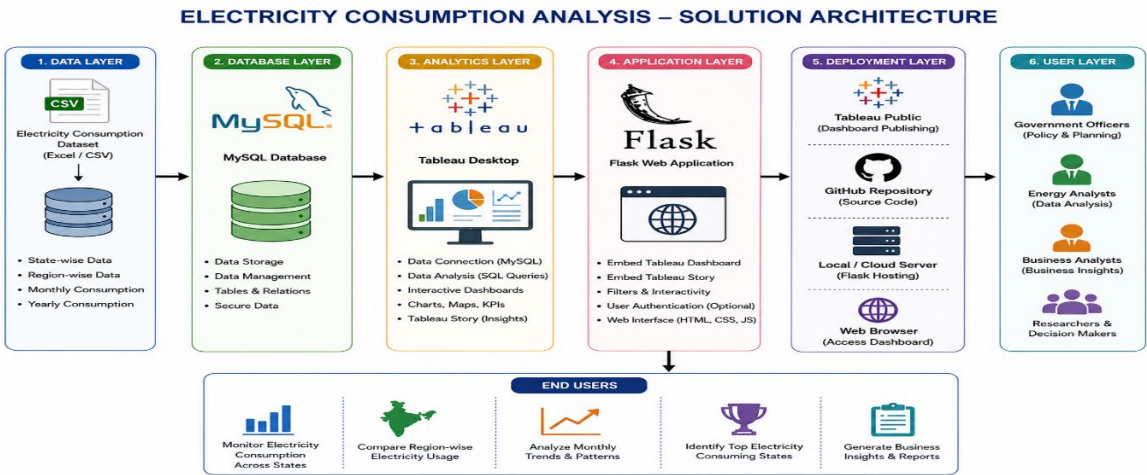
Solution Architecture

Solution Architecture is a systematic process that bridges the gap between business requirements and technology implementation. It defines how different technologies interact to solve the problem efficiently. The architecture of the **Electricity Consumption Analysis** project integrates data storage, database management, visualization, web deployment, and user interaction into a single analytical platform.

The primary goals of the solution architecture are to:

- Develop an efficient system for analyzing electricity consumption data.
- Define the overall structure and interaction between software components.
- Transform raw electricity consumption data into meaningful business insights.
- Provide interactive dashboards for better visualization and analysis.
- Deliver a web-based application that enables users to access electricity consumption reports easily.
- Support future enhancements such as AI-based demand forecasting and real-time electricity monitoring.

Solution Architecture Diagram



Architecture Layers

1. Data Layer

- Electricity Consumption Dataset (Excel/CSV)
- State-wise electricity usage records
- Region-wise electricity consumption data
- Monthly electricity usage information

2. Database Layer

- MySQL Database
- SQL Tables
- Data Storage
- SQL Query Processing

3. Analytics Layer

- Tableau Desktop
- Interactive Dashboards
- Tableau Story
- Business Insights

4. Application Layer

- Flask Framework
- Python
- HTML
- CSS
- JavaScript

5. Deployment Layer

- GitHub Repository
- Tableau Public
- Local Flask Server
- Web Browser

6. User Layer

- Government Officers
- Energy Analysts
- Business Analysts

- Researchers
- Decision Makers

Working Process

1. Import the electricity consumption dataset into MySQL.
2. Clean and validate the data.
3. Execute SQL queries to analyze electricity consumption.
4. Connect MySQL with Tableau Desktop.
5. Develop dashboards, maps, charts, and Tableau Story.
6. Publish dashboards to Tableau Public.
7. Integrate the published dashboard into a Flask web application.
8. Users access the application through a web browser.
9. Interactive dashboards provide business insights and support decision-making.

Advantages

- Efficient data storage and management.
- Interactive dashboards for better visualization.
- Faster data analysis and reporting.
- Easy access through a web application.
- Supports informed decision-making.
- Scalable for future enhancements.

Conclusion

The Solution Architecture provides a clear framework for developing the **Electricity Consumption Analysis** system. By integrating **MySQL**, **Tableau**, and **Flask**, the architecture ensures efficient data processing, interactive visualization, and easy accessibility. It enables users to analyze electricity consumption effectively and supports better energy planning and decision-making.